

CSCE 5290: NATURAL LANGUAGE PROCESSING

PROJECT 02: REPORT

GROUP NO.: 11

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COMPONENT 01: Prompting for Annotation Generation

1. Image 01: COCO_val2014_000000026671



Image ID = COCO_val2014_000000026671	Prompt A	Prompt B
Question Quality	5	5
Answer Correctness	4	5
Task Composition Quality	3	4
Question Confidence Reliability	5	5
Answer Confidence Reliability	4	5
Better Response	no	yes
Comment : The orange object might not be a carrot, it may be a stick, but no.of paws it is holding is definitely 2, so I think prompt B is best.		

2. Image 2: COCO_val2014_000000286717



Image ID = COCO_val2014_000000286717	Prompt A	Prompt B
Question Quality	5	5
Answer Correctness	4	5
Task Composition Quality	2	3
Question Confidence Reliability	5	5
Answer Confidence Reliability	5	5
Better Response	No	yes
Comment: both times the question is almost the same, but the quality like including specific color food items count has better answer than counting all the food items as many types, chutney, sauces, entries are there. But their confidences are spot on instead giving the highest scores.		

3. Image 3: COCO_val2014_000000100853



Image ID = COCO_val2014_000000100853	Prompt A	Prompt B
Question Quality	5	5
Answer Correctness	2	5
Task Composition Quality	4	3
Question Confidence Reliability	4	5
Answer Confidence Reliability	3	5
Better Response	no	yes
Comment: using minimal prompt a we the answer gave by model is long and its hard to guess, while using the context rich prompt we got single and correct answer.		

4. Image 4: COCO_val2014_000000436141



Image ID = COCO_val2014_000000436141	Prompt A	Prompt B
Question Quality	5	5
Answer Correctness	5	5
Task Composition Quality	3	4
Question Confidence Reliability	5	5
Answer Confidence Reliability	5	5
Better Response	Same, almost	yes
Comment: The only difference is the composition quality, as prompt b has captured 4 tasks while prompt a has captured only 3 tasks. Both prompts have asked almost similar questions but prompt b asked more things in the image than prompt A and both the prompts gave exact correct answer.		

5. Image 5: COCO_val2014_000000028463



Image ID = COCO_val2014_000000028463	Prompt A	Prompt B
Question Quality	5	5
Answer Correctness	4	5
Task Composition Quality	4	4
Question Confidence Reliability	5	5
Answer Confidence Reliability	5	5
Better Response	Almost same	Yes ,little bit better
Comment: Prompt B has given annotation that more direct answer 2, While prompt A has given that answer might not be correct in some cases. But the model has given less confidence to both question and answer , so the confidence reliability is high. Both prompts have given almost same level of annotation. But prompt b is better due to the direct answer type.		

6. Image 6: COCO_val2014_000000527464



Image ID = COCO_val2014_000000527464	Prompt A	Prompt B
Question Quality	5	5
Answer Correctness	5	5
Task Composition Quality	3	4
Question Confidence Reliability	5	5
Answer Confidence Reliability	5	5
Better Response	Almost the same	yes
Comment: The both prompts gave almost same level of annotations , its just that prompt B task compositions quality is high due to its 4 tasks, prompt A has given only 3 tasks.		

7. Image 7: COCO_val2014_000000092488



Image ID = COCO_val2014_000000092488	Prompt A	Prompt B
Question Quality	5	5
Answer Correctness	4	5
Task Composition Quality	2	4
Question Confidence Reliability	5	5
Answer Confidence Reliability	4	5
Better Response	no	yes
Comment: Although the prompt question is effect but answer may not always be same, some may spoon is metallic white , or some times silver like the answer. But in prompt B the answer is 1 as the No .of white ceramic cups behind the plate are just 1, the category cup is defined clearly, like don't count bowls. The tasks covered are higher for prompt B .		

8. Image 8: COCO_val2014_000000430469



Image ID = COCO_val2014_000000430469	Prompt A	Prompt B
Question Quality	5	5
Answer Correctness	5	5
Task Composition Quality	3	3
Question Confidence Reliability	5	5
Answer Confidence Reliability	5	4
Better Response	Yes	Almost the same
Comment: with both prompts the annotation quality is same , but in case of prompt B the answer given by model itself has or, Gold or yellow, then it should decrease confidence of answer to 4 but instead gave 5. So I deducted a point in the answer confidence reliability. Thus making prompt B a little worse than prompt A, but almost the same level.		

9. Image 9: COCO_val2014_000000005758



Image ID = COCO_val2014_000000005758	Prompt A	Prompt B
Question Quality	5	5
Answer Correctness	5	5
Task Composition Quality	3	3
Question Confidence Reliability	5	4
Answer Confidence Reliability	5	5
Better Response	yes	Almost the same.
Comment: both Prompts have almost the same level but the question wording or grammar is difficult to understand so, I have deducted the points in the question confidence reliability. So the prompt A is better a little due to complex questions in prompt B.		

10. Image 10: COCO_val2014_000000009041



Image ID = COCO_val2014_000000009041	Prompt A	Prompt B
Question Quality	5	4
Answer Correctness	4	5
Task Composition Quality	4	4
Question Confidence Reliability	5	5
Answer Confidence Reliability	4	5
Better Response	Almost the same	Yes
Comment: in prompt the answer given was white cat , it's a brown and white cat, this will give different answers so, I deducted one point each for answer correctness, and confidence reliability. But, the question in prompt B has become more generalized like was cat bigger or pigeon bigger so I have deducted 1 point for question quality.		

Question01:

Select the best AI-generated annotation from your evaluation (across both prompting strategies). Using this example, explain what makes a prompting strategy effective for generating high-quality annotations.

Answer:

The several annotations that are better, especially with the context rich prompting strategies. The best one I think is from 1st image, context rich prompt of my above 10 annotation images. Because it created question that is mixing recognition, location, counting and action tasks, in that question we must identify the dog ->recognition, the front paws-> location, the no. of paws-> count, the used for holding carrot-> action. A good prompting strategy is that it produces good quality annotations since it determines the way that the model will be used to produce outputs that are relevant, consistent and in line with the intent of the evaluation. The rich prompt example which I have chosen in the 1st image had some steps of reasoning in a single query. This demonstrates that the prompt was useful in attuning the model to produce a meaningful and well-constructed annotation rather than a question.

Neither did it produce any trivial product as it was a combination of such abilities as object identification, knowing where an object is, counting and action analysis. This demonstrates that the prompt came in handy in triggering the compositional thought, as opposed to the superficial observation. The necessity to include some thinking exercises was used to make sure that the annotation should be useful in further multimodal interpretation. The model was prompted to perform more than one multiple types of reasoning in a single annotation as opposed to: What animal is this? and how many paws can you see? The task schema like recognition, location, counting and action are comparable with a set of instructions which are used to instruct the model in analysing the image. By doing this, rather than coming up with random questions or questions with vague descriptions, the model is steered to direct its thinking with well-defined skills. This enhances quality of the annotation, and it is more acceptable as far as evaluation is concerned.

Meanwhile, structured output format as a question, answer, scores of confidences, and task composition allows all the annotations to be structured in a similar way. This minimizes the variability and no biased or ambiguous results. Due to these, the same results are achieved even to other pictures using the same prompt and when the results are similar, interpretable and in accordance to the objectives of the evaluation. This is particularly so in developing large datasets where the quality, consistency and understandability of argument directly affect the usefulness of annotations.

COMPONENT 02: Prompting to Correct Model Errors

1. Image: COCO_val2014_000000553968:



Question: What is the subject matter of the framed artwork hanging on the floral wallpaper?

Minimal Prompt: In the above image What is the subject matter of the framed artwork hanging on the floral wallpaper in max 3 words?

Model Response to Minimal Prompt: a person or figure in clothing

Explanation (if available): The framed artwork appears to depict a person or figure in clothing. I say this because within the frame, there is a central upright shape resembling a human figure with visible head-and-body proportions rather than a landscape or abstract pattern.

Final Revised Prompt: In the above image What is the two- or three-word subject matter of the framed artwork hanging on the floral wallpaper, Look only at the framed artwork above the bed. Try to identify the main object in the artwork using its visible shape of the which things are there, colors of the things, and context. The object appears to be a type of drink container for me. Do not guess person/animal unless clear, and give the reason why?

Model Response to Final Revised Prompt: Red wine bottle

Explanation (if available): the framed artwork appears to show a tall, narrow container with a dark neck/top and a lighter body or label, which matches the visible silhouette and proportions of a bottle-shaped drink container. The upright shape and centred presentation in the frame support that interpretation more than a person or animal. The color is mostly representing the wine as other alcohols are colourless.

Justification: The model failed because it's not considering the whole picture and floral part without zooming into portrait picture. The final revised prompt makes the model to focus solely on the framed portrait than full room. It also tells model to focus on shape of the objects, colors, and context of the surrounds with hint like drink container given, it identifies it be a wine bottle based on shape and color. Although it can any other drink bottle, the red and purple colors make it guess it as wine than other drinks. These descriptions make model to guess more accurate to our answer wine bottle rather than the previous guess animal or person which is more generic and not correct.

2. **Image:** COCO_val2014_000000404388



Question: What was the time?

Minimal Prompt: in the above image what was the time?

Model Response to Minimal Prompt: 12:00

Explanation by model (if available): The clock in the image appears to show 12:00 (as the both hands pointing straight up).

Final Revised Prompt: In the above image what is exact time shown in red clock on the wodden tool board, that has small hours hand on right and large minutes hand on left, the exact points are shown using the dots above the numbers, if the hand points to that dot its exactly an interval of 5 for minute hand, if not , its not a multiple of 5 for minute hand

Model Response to Final Revised Prompt: The exact time on the red clock appears to be 11:59

Explanation by model (if available): Large minute hand (left hand) is just slightly left of 12, not exactly on the top dot, so it is not exactly :00. It is about 59 minutes. Small hour hand (right hand) is just slightly before 12, which matches 11:59 rather than 12:00. The most accurate reading is 11:59 (about one minute before 12 o'clock).

Justification: It failed with the minimal prompt because the model is not guessing accurate answer and approximating and it is not correctly able to identify the hands of the clock. In the revised prompt we have provide the model on how to infer the time and which hands are left and right, which I think model found it difficult to differentiate and identified reversely in one of the iterations. The useful resourcefulness of the dots, and time intervals between the dots are also provided. These reasoning details helps the model with the visual details thus reducing ambiguity and making the model give a more accurate or exact answer 11.59 instead of approximation 12.00.

3. Image: COCO_val2014_000000195645



Question: How many kids are there?

Minimal Prompt: In the above picture, how many kids do you see?

Model Response to Minimal Prompt: 3

Explanation (if available): The 3 kids are located: Front left foreground — a toddler standing and facing the camera in a light-coloured hoodie. Centre background — a child in pink behind the bench, raising a drink/cup. Centre-right on the bench — a seated toddler wearing a hat and holding a cup.

Final Revised Prompt: From the image carefully analyse and identify the no. of children are there? First identify only clearly visible human faces. Count a face only if enough features are visible such as eyes, nose, mouth, or clear facial structure. Ignore faces that are blocked, blurry, turned away, or partial. Then, from those confirmed faces only, classify who is a child or adult using visible age cues like toddler or young child facial proportions. Do not guess age from clothing or body size alone. If age is uncertain, label as unknown and do not count as a child. Be conservative and exclude anything unclear.

Model Response to Final Revised Prompt: 2

Explanation (if available): Confirmed children: Front left toddler — clear full face with child features. Seated child on bench with hat — visible face with young child features. Not counted: Person in pink behind bench — face is partial and age unclear. All other clearly visible faces appear to be adults.

Justification: The failed because it considers the dress shadow the old man as a child due to poor visual cues, which is wrong to conder. But the revised prompt makes the model to count only the clearly visible faces with the defined features, instead of incorrectly identifying dress with shadows as face. This makes the reasoning more grounded with the basic rules of face recognition based on the evidence in image. Thus, splitting it into two tasks identifying faces and then classifying them as children makes the model to give more accurate answer.

4. **Image:** COCO_val2014_000000350959



Question: What color is the jacket of woman walking in the back along with man?

Minimal Prompt: in the above picture, what color is the jacket of woman walking in the back along with man?

Model Response to Minimal Prompt: black or grey

Explanation (if available): The person with long hair, likely woman has worn a black coat.

Final Revised Prompt: Analyse the picture carefully and first identify all clearly visible persons. Then identify each person is male or female, then relative to the identified male, estimate other person's relative depth using ground position, body size, overlap, and distance to background objects. Then find the person walking alongside him at the same depth level, meaning beside him laterally and not noticeably in front of or behind him. Do not choose based only on left-right closeness. If the person walking along is female give me her coat color.

Model Response to Final Revised Prompt: Light blue.

Explanation (if available): I used relative depth cues such as body size, vertical position on the ground, overlap, and distance to the background wall rather than just left-right spacing. The rightmost tall person appears male and is at a similar distance from the camera as the center person in the light blue jacket, while the smaller dark-coated person is farther ahead. Therefore, the person walking alongside the male is most likely the female in the light blue jacket.

Justification: The model makes a wrong answer with minimal prompt at because it distance metrics in 2d plane instead of 3d depth. So the refined prompt make model to consider visual depth estimation based on the size of the persons and distance from wall, like depth measurements and ground plane cues. This makes the model to avoid relying on simple left right distance between the image, which is a shock to me at first, that model measure distance in 2d plane instead of 3d. Thus, the revised prompt makes sure that the answer is correct and accurate while consider depth cues in a 3D space.

5. Image: COCO_val2014_000000268322



Question: Are there pillows on the bed?

Minimal Prompt: in the above picture of a room, Are there pillows on the bed?

Model Response to Minimal Prompt: No

Explanation (if available): No, there are no clearly visible pillows on the bed.

Final Revised Prompt: Look carefully at the above image of the room and locate the bed area first, then closely inspect the top section of the bed paying attention to corners, edges, raised shapes. Check through the partial visible objects too, do not depend on only fully visible objects, and confirm whether you are able to identify any pillows based on partial visible objects now.

Model Response to Final Revised Prompt: Yes — at least one pillow can be identified based on a partially visible object

Explanation (if available): I closely inspected the bed area first, especially the top/head section of the bed, including the corners, edges, raised areas under the netting, and partially visible objects. The Findings are as follows: The wooden carved headboard is clearly visible at the top centre of the bed. Most of the upper bed surface is covered by white bedding and partially obscured by the mosquito net. On the right side near the top edge of the bed, there appears to be a partially visible white pillow. Only part of its shape is visible, blending somewhat with the white sheets. On the left/top side, I do not see a clearly identifiable separate pillow shape.

Justification: The minimal prompt gives wrong answer at first as it not considers partial visible objects due previous trainings in the above image so, we had retrained the model with revised prompt to give correct answer. The revised prompt gives us less ambiguous answer which is accurate to say that there a pillow based on the specific recognition tasks dictated by us to perform. So now it checks every visual detail, such as the bed area, edges, and partially visible objects. Not only that we asked to do the reasoning of partial objects that are visible, but this also changes whole scenario to find the pillow that is only half visible with out ignoring it.

Analysis of Model Behavior:

In all these cases, the error in the model with the minimal prompt was often due to taking the most superficial, not enough understanding of the image rather than performing a precise visual check. On the very few prompts, it made a rough guess: identifying a bottle as a person, wrong guessing of time, increasing the number of children based on insufficient visual information, selecting a person based on shallow left-right proximity rather than depth cues, or not seeing a pillow due to its half-view visibility. Such mistakes may indicate that the model was operated based on a general pattern matching as opposed to a detailed examination of the specific area of identification in the question. When there are small ambiguous, occluded, or visually subtle details in the image, a short prompt provides too much space to think in a generic way, and, as such, the answer would be vague, approximate, or even incorrect.

The modified prompts were better due to the fact that they transformed the task to answer the question to a visual procedure task. They instructed the model on where to search, which clues to focus on and what sorts of guesses were to be avoided. That revisions have caused the reasoning to be more based on the image specific: the portrait prompts involved the interpretation of shape and object type, the clock prompt involved the interpretation of hands and the accuracy of minute, the children prompt involved the interpretation of faces relative to age, the jacket prompt involved the interpretation of relative depth, and the pillow prompt explicitly involved the checking of partial visibility. That is, the more successful prompts did not merely increase more words--they required superior cognition activities. Consequently, the responses were less ambiguous, more evidence-based, and more correct since the model was directed to look at the correct visual features rather than default assumptions.

Question 02:

In your examples, how can you determine whether the model's failure is due to a lack of specific capability (e.g., it counts incorrectly or it recognizes incorrectly), or due to poor prompting? (Specifically, if you are allowed to further query the AI system, describe how you would design follow up prompts to test your hypothesis. Your follow up prompts should aim to distinguish whether the issue is a limitation of the model's capability or a problem with how the question was asked.)

Answer:

In determining the cause of the failure in the model; i.e., whether it fails because it lacks capability or it fails because it fails because of the poor prompting we need to put the model to test using follow-up prompts that zero in on the point of failure. The failure must have been due to a bad prompting, and not to an inability, when the model responds correctly, after an elaborated, narrow or step-by-step prompting. When however, the model fails even after a lot of instruction, a lot of explanation and breaking down of the task, then it means that there is something amiss in the visual or reasoning ability of the model. In simple words, post-refinement improvement implies post-refinement sensitivity and recurrent failure implies ability constraints.

One is that where the model first responded 12:00 when it should have answered 11:59 in the clock picture, I would probe by asking more and more specific questions as follows: Name the minute hand only, Where is the minute hand pointing in relation to 12, Now name the hour hand. In the event that the model succeeds after such guided prompts, it was the root cause failure that was brought about by vague prompting. When it cannot distinguish the hands or when it cannot tell exactly when, then the deficiency comes under fine-grained visual perception. Similarly, in the counting image of children, I would first query it to determine all the faces that can be seen after which I would classify them as either child or adult. The problem was

planned well when the counting is enhanced by the decomposition; otherwise it indicates that there is no recognition ability.

The same testing method can be extended to recognition tests such as the framed art or pillow case. The model would be asked only to respond to visible shapes, colors and partially visible objects and then give the final answer, by asking a question. When it subsequently discovers the wine bottle or pillow, the earlier error was brought about by a question that was not specified well. The limitation is the ability to recognize objects in ambiguous or occluded conditions in case it fails despite the assistance of attention-directing prompts. This kind of controlled follow up prompting helps in decoupling whether or not the model is incapable of the task, or whether it simply was not asked to do it the right way.